Auto Chicken Counter Solution

1. Ultrasonic Sensor Solution

➔ Cost:
- Very cost effective solution. Easily available and interface-able.

➔ Power Consumption:
- Not very power hungry.
- Still not suitable for low battery operated situations compared to other options.

➔ Installation:
- Can be difficult to install.
- Requires proper mounting calculation to properly receive bounced waves.

➔ Benefits:
- Can be used in outside environments in sunlight or in dark environments.
- Not affected by dust or environmental conditions except water.

➔ Drawbacks:
- Can be easily confused when exposed to soft or non-uniform materials.

2. IR-Beam Sensor Solution

➔ Cost:
- Very cost effective and easy to use.

➔ Power Consumption:
- Very power efficient.
- Ideal for battery operated conditions.

➔ Installation:
- Easy to install and use.
- Two modules will be required to properly detect exit and enter situations of chicken.

- Will require some care while installing so that beams don't crosstalk with each other and give false data.

➔ Benefits:

- Easy availability and installation.
- Not affected by dust or environmental conditions except water.
- Fast response time.

➔ Drawbacks:

- Always the risk of cross talk between the two receivers and transmitters if they are located very close.
- Not suitable for use in sunlight conditions because Sun rays are also IR rays and can give false alarm at the receiver even when the chicken is blocking the IR beam.

3. LASER Light Solution:

➔ Cost:

- Very cost effective solution. Easily available and interface-able.

➔ Power Consumption:

- Highly power efficient.
- Ideal for low battery applications but consume more power compared to IR beams.

➔ Installation:

- Easy installation.
- Requires on-time proper adjustment of LASER light onto the receiver and it will work flawlessly for the rest of the time.

➔ Benefits:

- Highly directive and will properly detect chicken coming in and out with help of two modules.
- Not affected by dust or environmental conditions except water.

➔ Drawbacks:

- Not suitable for daylight conditions as the LASER may not reach the receiver if too much sunlight is present.
- LASER needs proper installation in order to be properly directive.

4. Force Sensitive Resistor Solution:

➔ Cost:
- Very cost effective solution. Easily available and interface-able.

➔ Power Consumption:
- Extremely low power as the sensing element is a resistor.
- Ideal for low power battery conditions.

➔ Installation:
- Can be difficult to install as it two FSRs will need to be placed at the entrance at some distance for detecting exit and enter.
- Harnessing to the main PCB will be a bit difficult compared to other options.

➔ Benefits:
- Can be used in outside environments in sunlight or in dark environments.
- Not affected by sunlight or dust or any environmental conditions.

➔ Drawbacks:
- Inconsistent output if the chicken pulls up its leg and then again places it on the sensor or moves around the sensor for some time, we may have multiple false alarms.
- Harnessing difficulties.

5. BLE Beacons Solution

➔ Cost:
- A bit costly solution will requires a small BLE strip to be worn by chicken all the time.

➔ Power Consumption:
- Extremely low power.
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➔ Installation:
- Easy installation but adds cost and complexity during development.
- Complete wireless so no installation complications.

➔ Benefits:

- Can be used in outside environments in sunlight or in dark environments.
- Not affected by sunlight or dust or any environmental conditions.
- Very accurate whenever the object is in range.

➔ Drawbacks:

- Difficult to distinguish chicken going in and out conditions.
- A bit complex compared to other solutions.